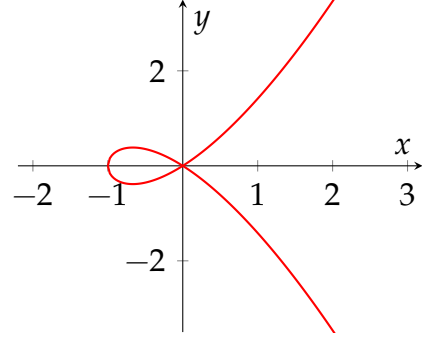


REES ALGEBRAS AND BLOWUP ALGEBRAS

PRISHITA DHARAMPAL

1. WHAT IS A BLOWUP?

We care about studying varieties and singularities are an obstruction to that. Intuitively, a singularity is a point in the variety where the tangent is not well-defined. For instance, in $\mathbb{A}^2$ the curve $y^2 = x^3 + x^2$ has node at the origin where two branches cross and we have two perfectly valid tangent directions, $y = x$ and $y = -x$. **TODO: add caption to the figure**



Blowups are a systematic tool for resolving singularities like this one. The idea is fairly straightforward, instead of picking one tangent direction at the singular point, we replace the point with a copy of $\mathbb{P}^1$, which parametrizes all lines through it simultaneously. The two branches then lift to the blowup as separate smooth curves, intersecting $\mathbb{P}^1$ at different points.

In general, the blowup of a variety $X \subset \mathbb{A}^n$ at a point replaces that point with a copy of $\mathbb{P}^{n-1}$, thus recording all the directions one can approach it from. So, it lives in $\mathbb{A}^n \times \mathbb{P}^{n-1}$.

Projective space $\mathbb{P}^n$ parametrizes lines through the origin in $\mathbb{A}^{n+1}$, i.e., a point $[p_0 : p_1 : \dots : p_{n-1}]$ represents a line through 0 and $(p_0, \dots, p_{n-1})$ in $\mathbb{A}^{n+1}$ with two points being equivalent if they differ by a non-zero scalar. Thus, a polynomial f only vanishes consistently at a projective point if $f(\lambda x) = 0$ whenever $f(x) = 0$, forcing f to be homogeneous. Therefore subvarieties of $\mathbb{P}^n$ are cut out by homogeneous ideals. For instance, given the (homogeneous coordinate) ring $k[x_0, \dots, x_n]$, in its associated projective space $\mathbb{P}^n$ we have

- (0) — corresponds to all of $\mathbb{P}^n$ **TODO: not accurate**
- (x_i) — a hyperplane $\{x_i = 0\} \cong \mathbb{P}^{n-1}$
- (x_i, x_j) — a codimension-2 linear subspace $\{x_i, x_j = 0\} \cong \mathbb{P}^{n-2}$
- (f) for homogeneous irreducible f of degree d - a hypersurface $\cong \mathbb{P}^{n-d}$.

2. GRADED RINGS AND MODULES

Since subvarieties of $\mathbb{P}^{n-1}$ are cut out by homogeneous ideals we need a framework to work with them. The polynomial ring $R = k[x_1, \dots, x_n]$ decomposes naturally by degree,

$$R = \bigoplus_{d \geq 0} k[x_1, \dots, x_n]_d$$

where $R_d = k[x_1, \dots, x_n]_d$ denotes homogeneous polynomials of degree d . Note that multiplication respects this decomposition,

$$R_d \cdot R_e \subseteq R_{d+e}$$

This ring $R = \bigoplus_{d \geq 0} R_d$ is called a graded ring. The polynomial ring has a natural grading by degree, but a general ring R doesn't. To extract a graded ring from R , we pick ideals $I_i \subset R$ such that the descending multiplicative chain holds,

$$R \supset I_1 \supset I_2 \supset I_3 \supset \dots$$

where $I_i I_j \subset I_{i+j}$. When the I_i are powers of a single ideal, this chain is called a I -adic filtration, and it allows us to measure "how deeply" an element lies in I . For our purposes, I is always the maximal ideal of a local Noetherian ring R .

From this filtration we can recover the associated graded ring $\text{gr}_I R$,

$$\text{gr}_I R := R/I \oplus I/I^2 \oplus \dots$$

Where the multiplication in $\text{gr}_I R$ is well-defined. For $a \in I^n/I^{n+1}$ and $b \in I^m/I^{m+1}$, we pick representatives $a' \in I^n$ and $b' \in I^m$ and define the product $ab \in I^{n+m}/I^{n+m+1}$ to be the image of $a'b'$. One can check that this is well-defined modulo I^{m+n+1} . We will now develop some theory around filtrations and graded rings that will be helpful later.

The idea of I -adic filtrations can be generalized to the case of modules.

Definition 2.1. Given an R -module M , and an ideal $I \subseteq R$ we get the I -adic filtration of the module,

$$M \supset IM \supset I^2 M \supset \dots$$

and the associated graded module,

$$\text{gr}_I M := M/IM \oplus IM/I^2 M \oplus I^2 M/I^3 M \oplus \dots$$

with multiplication defined as follows: for $a \in I^n/I^{n+1}$ and $b \in I^m M/I^{m+1} M$, we pick representatives $a' \in I^n$ and $b' \in I^m M$ and define the product $ab \in I^{n+m} M/I^{n+m+1} M$ to be the image of $a'b'$.

Remark 2.2. Note that intersecting the terms of the I -adic filtration of M with some submodule $M' \subset M$ sadly generally does not give us the I -adic filtration of M' .

The Artin-Rees lemma characterizes exactly the cases where this condition does hold. We will state it here and prove it after having developed the requisite theory.

Lemma 2.3 (Artin-Rees). Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R -modules. If

$$M = M_0 \supset M_1 \supset M_2 \supset \dots$$

is a I -stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \cdots$$

is also I -stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

Definition 2.4. Let R be a ring, $I \subseteq R$ an ideal, and M an R -module. A filtration $M = M_0 \supset M_1 \supset \cdots$ is called an I -filtration if $IM_n \subseteq M_{n+1}$ for all $n \geq 0$, and I -stable if there exists m such that $M_{m+i} = I^i M_m$ for all $i \geq 0$.

Notation. We write M_k to denote $I^k M$, and $\text{gr}_I R = \bigoplus_{k \geq 0} I^k / I^{k+1}$ and $\text{gr}_I M = \bigoplus_{k \geq 0} M_k / M_{k+1}$ for the associated graded ring and module respectively.

All ring maps $R \rightarrow \text{gr}_I R$ factor over the quotient R/I , since any such map must send I to positive degree elements and hence $I \subset \ker(\phi_0)$, where ϕ_0 denotes the degree 0 component,

$$\begin{array}{ccc} R & \xrightarrow{\phi} & \text{gr}_I R \\ \downarrow & \nearrow \exists! & \\ R/I & & \end{array}$$

However there is no interesting natural homomorphism from M to $\text{gr}_I M$, but there is an interesting natural set map called the initial form.

Definition 2.5. Let $f \in M$ and m be the greatest number such that $f \in M_m$. Then, the initial form of f , $\text{in}(f)$, is defined as,

$$\text{in}(f) = f \pmod{M_{m+1}} \in M_m / M_{m+1} \subset \text{gr}_I M$$

or if $f \in \bigcap_{m \geq 0} M_m$

$$\text{in}(f) = 0.$$

Definition 2.6. For a submodule $M' \subset M$ we define $\text{in}(M')$ to be the $\text{gr}_I R$ -submodule of $\text{gr}_I M$ generated by $\text{in}(f)$ for all $f \in M'$.

Remark 2.7. Unfortunately, $\text{in}(M')$ is generally not obtained as the submodule generated by the initial forms of the generators of M' .

For instance, take $R = k[x, y]$, $I = (x, y)$, and $J = (x^2 + y^3, xy) \subset R$ with the I -adic filtration. The initial forms of the generators are $\text{in}(x^2 + y^3) = x^2$ and $\text{in}(xy) = xy$, so the submodule generated by their initial forms is $(x^2, xy) \subset \text{gr}_I R \cong k[x, y]$. However, consider:

$$y \cdot (x^2 + y^3) - x \cdot (xy) = x^2 y + y^4 - x^2 y = y^4 \in J$$

so $\text{in}(y^4) = y^4 \in \text{in}(J)$. But $y^4 \notin (x^2, xy)$ in $k[x, y]$, since every element of (x^2, xy) has x as a factor while y^4 does not. Therefore $\text{in}(J) \supsetneq (\text{in}(x^2 + y^3), \text{in}(xy))$.

Proposition 2.8. Let R be a ring, $I \subset R$ be an ideal, and let M be a finitely generated R -module with I -filtration $\mathcal{J} : M = M_0 \supset M_1 \supset \cdots$ by finitely generated modules M_i . The filtration $\mathcal{J}$ is I -stable if and only if the $B_I R$ -module $B_{\mathcal{J}} M$ is finitely generated.

Proof.

($\Rightarrow$)

Assume $\mathcal{J}$ is I -stable. Then we have $M_{n+i} = I^i M_n$ for some n and all $i \geq 0$. That is, for all $k > n$, M_k is generated from M_n by multiplying powers of I . Since each $M_0, \dots, M_n$ is finitely-generated, the union of their generating sets gives a finite generating set for $B_{\mathcal{J}}M$.

($\Leftarrow$)

Assume $B_{\mathcal{J}}M$ is finitely generated. Since $B_{\mathcal{J}}M = \bigoplus_{n \geq 0} M_n$ is a direct sum, every element has only finitely many nonzero homogeneous components. Therefore each generator g_i is contained in $M_0 \oplus \dots \oplus M_{k_i}$ for some finite k_i , and taking $n = \max(k_1, \dots, k_r)$, all generators are contained in $M_0 \oplus \dots \oplus M_n$.

Each generator g_i decomposes as a finite sum of homogeneous pieces, and the $B_I R$ -submodule generated by all these homogeneous pieces contains the $B_I R$ -submodule generated by $\{g_i\}$, since each g_i is a sum of its homogeneous pieces. Since $\{g_i\}$ generates $B_{\mathcal{J}}M$, the homogeneous pieces also generate $B_{\mathcal{J}}M$. Hence we may assume $B_{\mathcal{J}}M$ is generated by homogeneous elements in $M_0, \dots, M_n$.

Since all generators live in degrees $\leq n$, any element of M_{n+i} is produced by multiplying a generator $m \in M_k$ with $k \leq n$ by some $a \in I^{n+i-k}$. Since $\mathcal{J}$ is an I -filtration, $I^{n-k}m \in M_n$, so $a \cdot m = I^i \cdot (I^{n-k}m) \in I^i M_n$. Therefore $M_{n+i} \subseteq I^i M_n$. The reverse inclusion $I^i M_n \subseteq M_{n+i}$ follows immediately from $\mathcal{J}$ being an I -filtration. Hence $M_{n+i} = I^i M_n$ for all $i \geq 0$, hence $\mathcal{J}$ is stable. **TODO: proofread** $\square$

Lemma 2.9 (Artin-Rees). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and $M' \subset M$ be finitely generated R -modules. If*

$$M = M_0 \supset M_1 \supset M_2 \supset \dots$$

is a I -stable filtration then the induced filtration

$$M' = M' \cap M_0 \supset M' \cap M_1 \supset M' \cap M_2 \supset \dots$$

is also I -stable, i.e., there exists m such that $I(M' \cap M_n) = M' \cap M_{n+1}$ for $n \geq m$.

Proof. Let

$$\mathcal{J}' : M' = M'_0 \supset M'_1 := M' \cap M_1 \supset M'_2 := M' \cap M_2 \supset \dots$$

be the induced filtration on M' . The module $B_{\mathcal{J}'}M' = \bigoplus_{n \geq 0} (M' \cap M_n)$ is naturally a graded $B_I R$ -submodule of $B_{\mathcal{J}}M$.

Since $\mathcal{J}$ is I -stable, $B_{\mathcal{J}}M$ is finitely generated as a $B_I R$ -module by Proposition 2.9. Since R is Noetherian and I is finitely generated, $B_I R = \bigoplus_{n \geq 0} I^n t^n$ is a finitely generated R -algebra and hence Noetherian by the Hilbert basis theorem. Therefore the submodule $B_{\mathcal{J}'}M'$ of the finitely generated $B_I R$ -module $B_{\mathcal{J}}M$ is itself finitely generated. Applying Proposition 2.9 again, $\mathcal{J}'$ is an I -stable filtration of M' . **TODO: Proofread** $\square$

Remark 2.10. *One of the important applications of associated graded constructions is when I is maximal ideal then $\text{gr}_I R$ is a graded algebra over the field R/I and if I is finitely-generated, $\text{gr}_I R$ is a finitely-generated algebra. Thus, gr_I allows us to turn arbitrary Noetherian rings into finitely-generated graded rings which are easier to study. **TODO: Why? tbh we loved Noetherian rings all term***

3. BLOW UP ALGEBRAS

Definition 3.1. Let R be a ring, and $I \subset R$ be an ideal. The **blowup algebra** of I in R is the R -algebra,

$$B_I R := \bigoplus_{n \geq 0} I^n t^n = R \oplus It \oplus I^2 t^2 \oplus \cdots \cong R[tI] \subset R[t]$$

where t is a formal variable tracking the degree of the ideal. The associated graded ring is obtained by quotienting by $IB_I R$:

$$B_I R / IB_I R = \text{gr}_I R = R/I \oplus I/I^2 \oplus \cdots$$

The blowup algebra is the algebraic structure behind the geometric operation of blowing up a subvariety. Suppose R is the coordinate k -algebra of an affine variety X , and $I \subset R$ is the ideal of a subvariety $Y \subset X$. Let $a_1, \dots, a_r$ be k -algebra generators for R and let $g_0, \dots, g_s$ be generators of I as an ideal of R .

Proposition 3.2. The blowup algebra $B_I R$ is a homomorphic image of $k[x_1, \dots, x_r, y_0, \dots, y_s]$.

Proof. Define a ring homomorphism

$$\phi : k[x_1, \dots, x_r, y_0, \dots, y_s] \rightarrow B_I R, \quad x_i \mapsto a_i, \quad y_j \mapsto g_j t$$

This is surjective since any element of $B_I R$ is a finite sum of terms of the form $f \cdot g t^n$ with $f \in R$ and $g \in I^n$, and since f is a polynomial in the a_i and g is a polynomial in the g_j , every such term is in the image of ϕ . By the first isomorphism theorem:

$$B_I R \cong k[x_1, \dots, x_r, y_0, \dots, y_s] / \ker(\phi)$$

Since $B_I R$ is a graded ring and ϕ respects the grading by sending the x_i to degree 0 elements and the y_j to degree 1 elements, the kernel is a homogeneous ideal. $\square$

$k[x_1, \dots, x_r]$ is the coordinate ring of $\mathbb{A}^r$, and $k[y_0, \dots, y_s]$ is the homogeneous coordinate ring of $\mathbb{P}^s$. So $k[x_1, \dots, x_r, y_0, \dots, y_s]$ is the coordinate ring of the product $\mathbb{A}^r \times \mathbb{A}^{s+1}$, and quotienting by a homogeneous ideal in the y_j variables makes the vanishing scale-invariant in the y_j and hence cuts out a subvariety that is projective in the y -directions,

$$Z = \mathbb{V}(\ker \phi) \subset \mathbb{A}^r \times \mathbb{P}^s$$

The projection $\pi : \mathbb{A}^r \times \mathbb{P}^s \rightarrow \mathbb{A}^r$ restricts to a map $Z \rightarrow X$ which is an isomorphism away from the preimage of Y . The set Z is called the **blowup** of Y in X , and the preimage $\pi^{-1}(Y) \subset Z$ is called the **exceptional set**.

TODO: Algebraically, the exceptional set corresponds to:

$$B_I R / IB_I R = \text{gr}_I R$$

The exceptional set is the projective variety associated to $\text{gr}_I R$. Recall that $\text{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$ is a graded ring, and its homogeneous prime ideals not containing all of $(\text{gr}_I R)_{>0}$ correspond to points of a projective variety, exactly as homogeneous prime ideals of $k[x_0, \dots, x_s]$ correspond to points of $\mathbb{P}^s$. This variety parametrizes all limiting directions of approach to Y inside X — the degree 1 piece I/I^2 is the space of directions,

and taking its projectivization replaces Y with the set of all lines through it, one for each limiting direction of a secant line to X through Y .

We are now ready to revisit the example we looked at at the start of this note more formally,

Example: Blowing up the node of $y^2 = x^3 - x^2$. The curve $y^2 = x^3 - x^2$ has a node at the origin. The Jacobian $(-3x^2 + 2x, 2y)$ vanishes at $(0, 0)$, confirming it is singular.

We blow up $Y = \{0\}$ inside $X = \mathbb{V}(y^2 - x^3 + x^2)$. The coordinate ring is $R = k[x, y] / \langle y^2 - x^3 + x^2 \rangle$ and the maximal ideal of the origin is $I = \langle x, y \rangle$ (by Nullstellensatz).

The generators of R as a k -algebra are $\{x, y\}$ and the generators of I are also $\{x, y\}$. Relabeling the generators of R as x_1, x_2 and the generators of I as y_0, y_1 , the blowup algebra is a homomorphic image of $k[x_1, x_2, y_0, y_1]$ by the surjection,

$$\phi : x_1 \mapsto x, \quad x_2 \mapsto y, \quad y_0 \mapsto xt, \quad y_1 \mapsto yt$$

The kernel is generated by the curve relation $x_2^2 - x_1^3 + x_1^2$ and the relation $x_1y_1 - x_2y_0$, which lies in $\ker(\phi)$ since $\phi(x_1y_1 - x_2y_0) = x \cdot yt - y \cdot xt = 0$. Since this relation is homogeneous of degree 1 in y_0, y_1 , the blowup cuts out an algebraic subset $Z \subset \mathbb{A}^2 \times \mathbb{P}^1$.

4. TANGENT CONE

Theorem 4.1 (Krull's Intersection Theorem). *Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R -module. There exists an element $r \in I$ such that*

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0$$

Proof. Let $N = \bigcap_{k=1}^{\infty} I^k M$. We want to find $r \in I$ such that $(1 - r)N = 0$. Consider the I -adic filtration on M ,

$$M \supset IM \supset I^2 M \supset \dots$$

and the induced filtration on N ,

$$N \supset N \cap IM \supset N \cap I^2 M \supset \dots$$

By Artin-Rees applied to $N \subset M$, there exists m such that for all $n \geq m$,

$$N \cap I^n M = I^{n-m}(N \cap I^m M)$$

In particular,

$$N \cap I^{m+1} M = I(N \cap I^m M)$$

But since N is the intersection of all $I^k M$ we have $N \subseteq I^{m+1} M$, so $N \cap I^{m+1} M = N$. Similarly $N \subseteq I^m M$, so $N \cap I^m M = N$. Therefore:

$$N = IN$$

Since R is Noetherian and M is finitely generated, N is a finitely generated R -module. Applying Nakayama's lemma to $N = IN$: there exists $r \in I$ such that $(1 - r)N = 0$. $\square$

Corollary 4.1.1. *If R is a domain or $(R, \mathfrak{m})$ is a local Noetherian ring, and $I \subsetneq R$ then*

$$\bigcap_{k=1}^{\infty} I^k = 0$$

Proof. Let $M = R$. Since I is a proper ideal, there exists some $r \in I$ such that $r \neq 1$, so $1 - r \neq 0$. If R is a domain, we are done. If R is a local ring, $r \in I \subseteq \mathfrak{m}$ so $1 - r \notin \mathfrak{m}$, hence $1 - r$ is a unit and must be non-zero. Hence, $\bigcap_{k=1}^{\infty} I^k = 0$. $\square$

Corollary 4.1.2. *If R is a Noetherian local ring and $I \subsetneq R$ is an ideal then if $\text{gr}_I R$ is a domain, so is R .*

Proof. If $fg = 0 \in R$ then $\text{in}(f)\text{in}(g) = 0 \in \text{gr}_I R$. Since $\text{gr}_I R$ is a domain, $\text{in}(f)$ or $\text{in}(g)$ is zero. Without loss of generality, assume $\text{in}(f) = 0$. Then by definition, $f \in \bigcap_{k=1}^{\infty} I^k$ and by KIT we know that $\bigcap_{k=1}^{\infty} I^k = 0$. Hence $f = 0$. Thus, R is a domain. $\square$

Definition 4.2. *A line through 0 in $\mathbb{A}^r$ is determined by its direction (which corresponds to a point in $\mathbb{P}^{r-1}$). More precisely, the line through 0 and $p = (p_1, \dots, p_r)$ corresponds to $[p_1 : \dots : p_r] \in \mathbb{P}^{r-1}$. So a sequence of secant lines ℓ_{p_n} at a point $X \subset \mathbb{A}^r$ is really a sequence of points $[p_n] \in \mathbb{P}^{r-1}$. A **limiting position** is a limit point of this sequence:*

$$\ell = \lim_{n \rightarrow \infty} [p_n] \in \mathbb{P}^{r-1}$$

This is well-defined because $\mathbb{P}^{r-1}$ is compact (over $\mathbb{C}$), so every sequence has a convergent subsequence.

The correspondence between the exceptional set in the blow up and the associated graded ring gives us a beautiful geometric consequence. Let $R = k[x_1, \dots, x_r]/J$ and $I = (x_1, \dots, x_r)$ where k is algebraically closed and $J \subset I$, also let X be the algebraic subset of $\mathbb{A}^r$ cut out by J ,

$$X = \mathbb{V}(J) \subset \mathbb{A}^r \text{ so } 0 \in X$$

Then we define the tangent cone of X at 0 as the cone composed of all lines that are the limiting positions of secant lines to X passing through the point 0. Alternatively, the tangent cone at 0 is then the closure of the set of directions of points on X , together with the actual lines they represent in $\mathbb{A}^r$.

$$TC_0(X) = \overline{\{[p] \in \mathbb{P}^{r-1} : p \in X \setminus \{0\}\}} \subset \mathbb{P}^{r-1}$$

Since the closure captures limiting directions that arise as limits of secant lines but may not themselves be a point of $X \setminus \{0\}$.

5. CONCLUSION

TODO: Talk about

6. WORKSHEET

TODO: Add figure and theorem

Theorem 6.1. Let $R = k[x_1, \dots, x_r]/J$ with $I = (x_1, \dots, x_r)$ and k algebraically closed. The tangent cone $TC_0(X)$ to $X = \mathbb{V}(J) \subset \mathbb{A}^r$ at the origin is the algebraic subset:

$$TC_0(X) = \mathbb{V}(\text{in}_I(J)) \subset \mathbb{A}^r$$

where $\text{in}_I(J) = \langle \text{in}(f) : f \in J \rangle \subset k[x_1, \dots, x_r]$ is the ideal generated by the initial forms of all elements of J . Moreover, the coordinate ring of the tangent cone is:

$$k[x_1, \dots, x_r]/\text{in}_I(J) \cong \text{gr}_I R$$

Proof. We sketch the key steps; for a complete proof see Harris

First, $\text{in}_I(J)$ is a homogeneous ideal since each $\text{in}(f)$ is homogeneous by definition, so $\mathbb{V}(\text{in}_I(J))$ is indeed a cone in $\mathbb{A}^r$.

For the geometric identification: a direction $[v] \in \mathbb{P}^{r-1}$ lies in $TC_0(X)$ if and only if v is a limit of directions $[p_n]$ with $p_n \in X \setminus \{0\}$ and $p_n \rightarrow 0$. Writing $p_n = t_n v_n$ with $t_n \rightarrow 0$ and $v_n \rightarrow v$, and expanding $f(t_n v_n) = 0$ for $f \in J$, the leading order term gives $\text{in}(f)(v) = 0$. Since this holds for all $f \in J$, we get $v \in \mathbb{V}(\text{in}_I(J))$. The converse holds by a similar limiting argument.

For the ring isomorphism, note that the natural map:

$$k[x_1, \dots, x_r] \twoheadrightarrow \text{gr}_I R = \bigoplus_{n \geq 0} I^n / I^{n+1}$$

sending x_i to its class in I/I^2 is surjective with kernel $\text{in}_I(J)$, giving $\text{gr}_I R \cong k[x_1, \dots, x_r]/\text{in}_I(J)$. □

Example 6.1. For the nodal curve $y^2 = x^3 - x^2$, the lowest degree part of $f = y^2 - x^3 + x^2$ is $\text{in}_I(f) = y^2 - x^2 = (y - x)(y + x)$. So the tangent cone is $\mathbb{V}(y^2 - x^2) \subset \mathbb{A}^2$, the union of the two lines $y = x$ and $y = -x$, confirming the two tangent directions at the node. The coordinate ring is $k[x, y]/(y^2 - x^2) \cong \text{gr}_I R$.

Problem 1.

- (1) Motivation
- (2) KIT
- (3) Blow Ups, Blow Up Algebras, Exceptional Sets
- (4) Limiting Points and Tangent Cones
- (5) Flatness
- (6) Projective and Free Resolutions
- (7) 'Criteria for Flatness'
- (8) Rees Algebras

- (1) **TODO: Examples**
- (2) What makes a good family? Flatness!

- Algebraically, this means that when the family is represented by a R -algebra S , then S is flat as a R -module.
 - Geometrically this means that when the family is represented by a morphism $\phi : X \rightarrow B$, then for each point $x \in X$ there is an affine neighborhood U of x and an affine neighborhood of $\phi(x)$ such that ϕ restricts to a map $U \rightarrow V$ and the corresponding map of coordinate rings $K[V] \rightarrow K[U]$ makes $K[U]$ into a flat $K[V]$ -module.
- (3) Free resolution helps "standardize" the sequence so tor can give us a measure of the failure of exactness **TODO: examples and definitions**
- (4) **TODO: Criteria for Flatness.**
- (5) **TODO: Rees Algebras as a flat family.**